

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicentre, Randomised, Double-blind, Parallel-group, Placebo-controlled Study to Evaluate the Efficacy and Safety of Tozorakimab (MEDI3506) in Patients Hospitalised for Viral Lung Infection Requiring Supplemental Oxygen (TILIA) **Short Title/Acronym:** TILIA **Protocol Number:** D9185C00001 **NCT Number:** NCT05624450 **Sponsor Name:** AstraZeneca **Investigational Product:** Tozorakimab (MEDI3506) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of tozorakimab, as an add-on to Standard of Care (SoC) in patients with viral lung infection requiring supplemental oxygen, on the prevention of death or progression to invasive mechanical ventilation (IMV) or extracorporeal membrane oxygenation (ECMO). **Secondary Objectives:** To evaluate the use of baseline serum IL-33/sST2 complex levels (U/mL) to predict treatment response with tozorakimab versus placebo, and to assess the overall safety and tolerability of the drug.

2.2 Background & Rationale Severe viral lung infections can potentially lead to Acute Respiratory Distress Syndrome (ARDS), a life-threatening lung injury often treated with invasive mechanical ventilation (IMV). Tozorakimab is an antibody designed to target and block interleukin-33 (IL-33), a protein that plays a key role in the lung's inflammatory cascade. By inhibiting excess IL-33 signaling, it is hypothesized that the drug will suppress inflammation and prevent the progression of viral lung infections to severe lung injury and unfavorable outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1 Parallel Assignment

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 2,870 evaluable patients **Number of Sites:** Multicentre (Global) **Estimated Study Start:** 13-Dec-2022 **Estimated Primary Completion:** 19-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult participants ≥ 18 years old at the time of signing the informed consent form. 2. Patients hospitalised with viral lung infection. 3. Hypoxaemia requiring treatment with supplemental O₂. Hypoxaemia is defined as: $SpO_2 \leq 90\%$ OR $SpO_2 \leq 92\%$ AND one or both of the following: radiographic infiltrates by CXR/CT compatible with viral lung infection per investigator judgement; use of accessory muscles of respiration or respiratory rate (RR) > 22 . 4. Patient remains hypoxaemic at randomisation requiring treatment with supplemental oxygen.

4.2 Exclusion Criteria 1. Known fungal or parasitic lung infection, aspiration lung infection, lung abscess, or evidence of septic shock. (Bacterial co-infection is allowed, unless bacterial infection defines the severity of the condition). 2. Hypoxaemia caused primarily by extrapulmonary insult or by lung injury of non-infective aetiology. 3. Ongoing IMV/ECMO at randomisation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose on Day 1. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Single administration.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of Care (SoC) for viral lung infection and supplemental oxygen therapy. **Prohibited:** Concurrent investigational agents or therapies that conflict with the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -14 to 0	Informed Consent, Medical History, Vital Signs, SpO ₂ / Imaging
2	Day 0	Randomization, Single IV Dose (Tozorakimab or Placebo)
3	Week 1 to 4	Safety Labs, Efficacy Assessment, Continuous Monitoring for IMV/ECMO
4	Follow-up	Final Vital Signs, Exit Interview, AE/SAE Collection

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Prevention of death or progression to Invasive Mechanical Ventilation (IMV) or Extracorporeal Membrane Oxygenation (ECMO). **Measurement Method:** Clinical observation, tracking of survival, and mechanical respiratory support requirements in the hospital setting.

7.2 Secondary & Exploratory Endpoints * **Endpoint 1:** Evaluation of baseline serum interleukin-33/soluble Suppression of Tumorigenicity 2 (IL-33/sST2) complex levels (U/mL) relative to primary endpoint to predict treatment response. * **Endpoint 2:** General safety and tolerability profile (AEs and SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard intensive clinical evaluation and patient monitoring during the hospitalization and follow-up phase. **Serious Adverse Events (SAEs):** Reported within standard 24-hour reporting timelines, particularly concerning progression to ARDS, IMV/ECMO, or death. **Data Monitoring Committee (DMC):** External Independent Data Monitoring Committee (IDMC) implemented for safety oversight during the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for adverse events. **Sample Size Justification:** Target enrollment of $N = 2,870$ appropriately powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect a statistically significant reduction in disease progression or mortality. **Handling of Missing Data:** Standard survival analysis methods including right-censoring for time-to-event outcomes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring for hospitalized acute care settings. **Audit Trail:** Electronic Case Report Form (eCRF) locking, source data verification, and data clarification mechanisms.

11. Study Identifiers & Governance

Primary ID: D9185C00001 **Registry Number:** NCT05624450 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** TILIA **Official Regulatory Title:** A Phase III, Multicentre, Randomised, Double-blind, Parallel-group, Placebo-controlled Study to Evaluate the Efficacy and Safety of Tozorakimab (MEDI3506) in Patients Hospitalised for Viral Lung Infection Requiring Supplemental Oxygen (TILIA)

12. Clinical Framework (Study Specific)

Primary Condition: Viral Lung Infection Requiring Supplemental Oxygen **Diagnosis Threshold:** $SpO_2 \leq 90\%$ OR $SpO_2 \leq 92\%$ with signs of respiratory distress (RR > 22 or use of accessory muscles) **Medical Methodology:** Standard of Care (SoC) plus investigational anti-IL-33 biologic **Optimization Target:** Prevention of disease progression to IMV/ECMO or death

13. Study Procedures & Methodology

Management Pathway: Hospitalized acute care pathway for viral lung infections. **Education Protocol (HCP):** Protocol-specific training for single IV dose administration and IMV/ECMO criteria. **Education Protocol (Patient):** Informed consent process and hospitalization procedures. **Quality Audit Mechanism:** Clinical trial monitoring and real-time review of progression events.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up through primary completion and protocol-defined safety observation period. **Onsite Visit Cadence:** Daily acute care monitoring while hospitalized. **Remote Monitoring Frequency:** Standard post-discharge follow-up calls per protocol. **Checklist Review Interval:** Continuous throughout the acute hospitalization phase.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				